

Deep Learning

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- Simple Cross Entropy

$$L(w) = \frac{1}{n} \sum_{i=1}^n \left(-y_i^T \log f_w(x_i) \right).$$

Balanced Cross Entropy

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$$L(w) = \frac{1}{n} \sum_{i=1}^n \left(-y_i^T \log f_w(x_i) \right).$$

- Balanced Cross Entropy

$$L(w) = \frac{1}{n} \sum_{i=1}^n \left(-(\alpha * y_i)^T \log f_w(x_i) \right).$$

Where $\alpha^T = [\alpha_1, \dots, \alpha_k]$, $\alpha_i \in (0, 1)$ and $\sum_{i=1}^k \alpha_i = 1$.

1 Basic Recurrent Neural Networks

- What is sequential data?

Sequential Data

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 - Time series

Sequential Data

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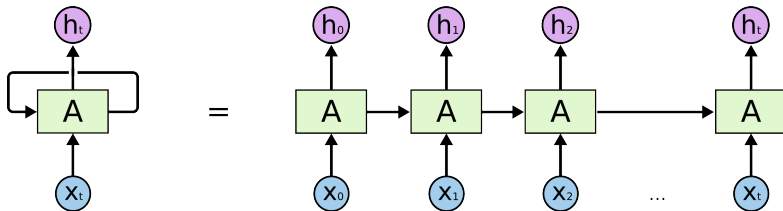
Sequential Data

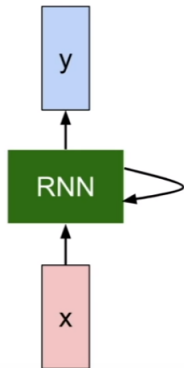
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- Why don't to use known networks?
 - Inputs, outputs can be different lengths in different examples.
 - Doesn't share features learned across different positions of sequence.

Basic RNNs





$$h_t = f_W(h_{t-1}, x_t)$$



$$h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t)$$

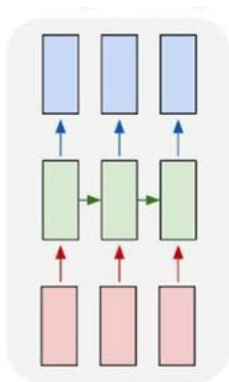
$$y_t = W_{hy}h_t$$

Different Types of RNNs

- Many to many

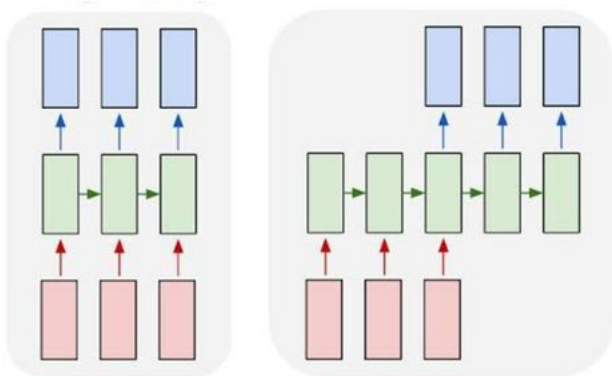
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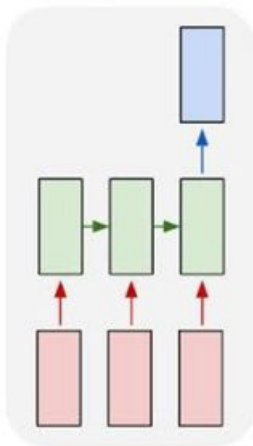


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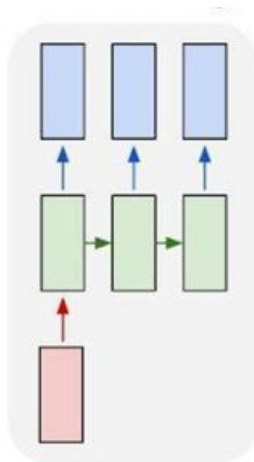


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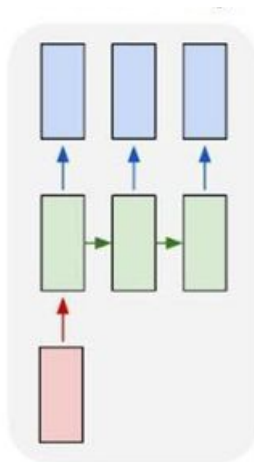
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Problem of Long Term Dependencies

